Replicating Abadie, Diamond & Hainmueller (2010)

Synthetic control and California's Proposition 99

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STAT 5200: Applied Econometrics Replication presentation

The paper and why I chose it

Abadie, Diamond & Hainmueller (2010), *JASA* 105(490): 493–505.

Question. What happened to cigarette sales in California because of Proposition 99, a 1988 tobacco control program?

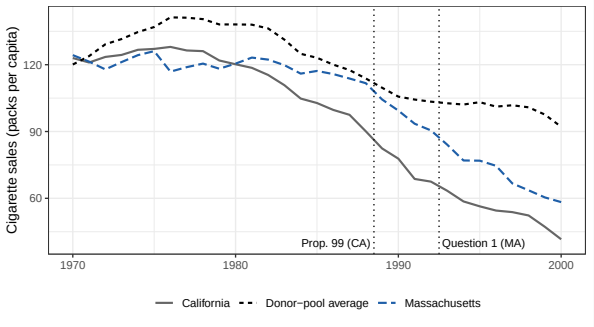
Why it is hard. One treated state. No other state looks like California before 1988, so a simple DiD against "the rest of the US" is not credible.

Contribution. The synthetic control method: build the comparison unit as a data-driven weighted average of untreated states, and do inference by placebo permutation.

Link to my project. Massachusetts passed an almost identical program in 1992 (Question 1). My project applies the same design, data and code to Massachusetts and compares it with DiD and synthetic DiD.

Setting: Proposition 99

- Passed November 1988, in effect January 1989
- Cigarette excise tax +25 cents per pack
- Revenue earmarked for health and anti-smoking education: media campaigns, school programs, local clean-air ordinances
- Over $100 million per year in anti-tobacco projects at first



Per-capita sales. California was already declining faster than the donor average before 1989.

Method: synthetic control

States $j = 1$ (California) and $j = 2, \ldots, J+1$ ($J = 38$ donors); last untreated year $T_0 = 1988$. The effect in year $t > T_0$ is

$$\alpha_{1t} = Y_{1t}^I - Y_{1t}^N, \qquad \hat{Y}_{1t}^N = \sum_{j=2}^{J+1} w_j^* Y_{jt}, \quad w_j \geq 0, \ \sum_j w_j = 1.$$

Weights match pre-1989 predictors X_1 (California) and X_0 (donors):

$$W^*(V) = \arg\min_W \ (X_1 - X_0 W)' V (X_1 - X_0 W),$$

with diagonal V chosen to minimise the pre-1989 MSPE of sales.

Why it works. If sales follow a factor model, matching a long pre-period of outcomes also matches the unobserved factor loadings. Weights are non-negative and sum to one, so there is no extrapolation.

Inference. Reassign treatment to each donor in turn. Rank California's post/pre MSPE ratio among all 39 states.

Data and design choices in the paper

- **Outcome:** per-capita cigarette sales in packs, 1970–2000, from tax revenue (Orzechowski & Walker, *The Tax Burden on Tobacco*)
- **Predictors:** log income per capita, retail price, share aged 15–24 (averaged 1980–88); beer per capita (1984–88); sales in 1975, 1980 and 1988
- **Donor pool, 38 states.** Dropped:
  - states with large tobacco programs in 1989–2000: MA, AZ, OR, FL
  - states that raised cigarette taxes by 50 cents or more: AK, HI, MD, MI, NJ, NY, WA
  - District of Columbia
- **Sample ends in 2000**: after that many states adopted anti-tobacco measures (e.g. the 1998 Master Settlement Agreement)

Paper's results

- Synthetic California = Utah 0.334, Nevada 0.234, Montana 0.199, Colorado 0.164, Connecticut 0.069
- Pre-1989 fit is close: MSPE about 3
- By 2000, sales were **about 26 packs per capita lower** than synthetic California
- Average over 1989–2000: almost 20 packs lower, **about 25%**
- California's post/pre MSPE ratio is about 130, the largest of the 39 states: **permutation** $p = 1/39 = 0.026$

My replication

Goal: rebuild the data from public sources and reproduce the analysis with the original estimator, not with the authors' data file.

Ingredient	Source used here
Sales and price	CDC STATE System, *Tax Burden on Tobacco 1970–2019*
Income	BEA SAINC1, per capita personal income (nominal, logged)
Share aged 15–24	NCI SEER population by age
Beer	NIAAA apparent consumption, gallons per resident 14+
Estimator	R package `Synth` (same as the authors' software)

- Same donor pool, same predictors and averaging windows
- Pipeline: `Rscript run_all.R` downloads the raw data and regenerates every table and figure
- Code: `https://github.com/yang-qs/stat5200-project`

Predictor balance: paper vs replication

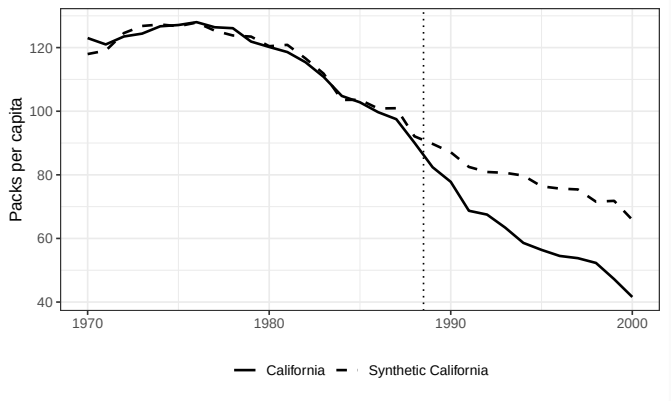
	ADH (2010)		Replication	
	California	Synthetic	California	Synthetic
Ln(income per capita)	10.08	9.86	9.65	9.49
Retail price	89.42	89.41	96.57	97.97
Share aged 15–24 (%)	17.40	17.40	17.32	17.04
Beer per capita (gal.)	24.28	24.20	29.79	30.03
Packs per capita, 1975	127.10	126.99	127.10	126.73
Packs per capita, 1980	120.20	120.43	120.20	120.44
Packs per capita, 1988	90.10	91.62	90.10	92.11

Prices in cents. Lagged sales match the paper to the decimal. Covariate levels differ because the definitions differ: nominal personal income vs. 1997 dollars, beer per resident 14+ vs. per capita, a revised price series. Each state's predictors use the same definition, so only relative positions matter for the weights.

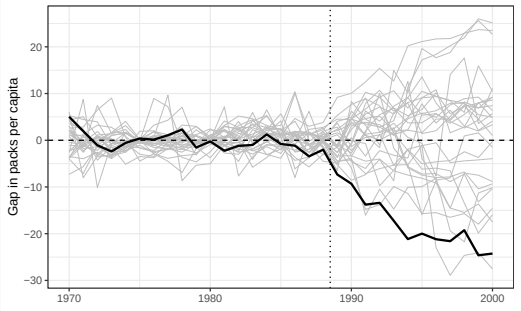
Synthetic California: weights and paths

State	ADH (2010)	Replication
UT	0.334	0.367
CO	0.164	0.316
NV	0.234	0.237
MT	0.199	–
CT	0.069	0.077

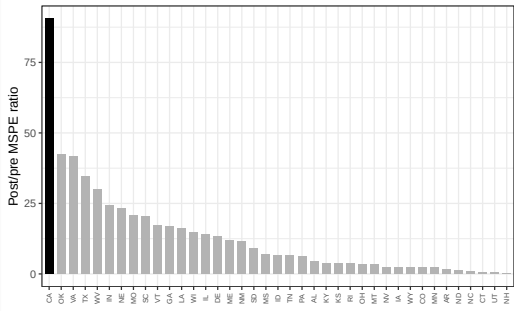
Four of the paper's five donors come back. Montana drops out and Colorado's weight roughly doubles.



Inference: placebos in space



Gaps for California (black) and the placebo states (grey). Placebos with pre-1989 MSPE above 5 times California's are dropped, as in the paper's Figure 6.



Post/pre MSPE ratio (cf. paper's Figure 8). California ranks first of 39.

Headline numbers

	ADH (2010)	Replication
Pre-1989 MSPE	about 3	3.8
Gap in 2000 (packs)	about -26	-24.3
Average gap 1989–2000 (packs)	almost -20	-17.8
Average gap, % of synthetic	about -25%	-23%
Post/pre MSPE ratio	about 130	90
Rank of California among 39	1	1
Permutation p-value	$1/39 = 0.026$	0.026

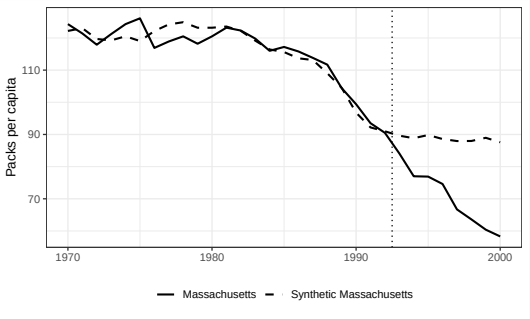
The qualitative result and the inference replicate exactly. The point estimate is about 10% smaller, and the fit is a bit worse, so the MSPE ratio is lower. The p-value is identical because California still has the largest ratio.

Lessons from the replication

1. **Data vintage matters, but less than I expected.** Lagged sales match the paper exactly, while the price series and every covariate differ in definition or source. The 2000 gap moved by about 2 packs.
2. **Read the paper closely.** Table 1 says "Ln(GDP per capita)" in 1997 dollars; the text and data appendix say log personal income. I followed the appendix.
3. **Weights are less stable than the counterfactual.** Small data changes dropped Montana and doubled Colorado, but the synthetic path and the gap barely moved. Interpret the path, not individual weights.
4. **Software is fragile.** `Synth`'s optimizer failed on one placebo state in the Massachusetts design; I added a retry with looser tolerances.
5. **Fiscal vs. calendar years.** Sales are by fiscal year (ending June 30). Following the paper, the fiscal year containing the January tax increase is the first treated year.

How this informs my project: Massachusetts 1992

- Question 1: tax 26 $\rightarrow$ 51 cents on 1 January 1993, revenue funds a tobacco control program
- Same data, donor pool, predictors and code as the replication, so differences in answers are not differences in method
- Synthetic MA = CT 0.49, DE 0.29, UT 0.15, RI 0.07
- Average gap 1993–2000: -18.5 packs (21%); permutation $p = 0.077$
- Add TWFE DiD and synthetic DiD (Arkhangelsky et al. 2021) to see how much the estimator matters



Takeaways

- The main result of ADH (2010) replicates from public data: a gap of 24 packs by 2000 (paper: 26), and California ranks first of 39 placebos ($p = 0.026$).
- The method is transparent about its comparison unit: the weights and pre-period fit can be inspected and criticised.
- With one treated unit, inference rests on placebo permutations, and the smallest attainable p-value is $1/(J+1)$.
- Open issues the paper shares with my project: sales are not smoking (cross-border purchases), and the treatment bundles a tax with a program.

Thank you. Code: `https://github.com/yang-qs/stat5200-project`

References

Abadie, A., Diamond, A., and Hainmueller, J. (2010). Synthetic control methods for comparative case studies: Estimating the effect of California's tobacco control program. *Journal of the American Statistical Association*, 105(490):493–505.

Abadie, A. and Gardeazabal, J. (2003). The economic costs of conflict: A case study of the Basque Country. *American Economic Review*, 93(1):113–132.

Arkhangelsky, D., Athey, S., Hirshberg, D. A., Imbens, G. W., and Wager, S. (2021). Synthetic difference-in-differences. *American Economic Review*, 111(12):4088–4118.

Orzechowski and Walker (2019). *The Tax Burden on Tobacco, 1970–2019*. CDC STATE System, dataset `7nwe-3aj9`.